Class Field Theory VIII.6.13

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Original statement

Source `CFT.pdf`*, physical page 265.*

LEMMA 6.13 *Let T be finite set of real or finite primes of K, and let $b \in K^\times$. If T has an even number of elements and b does not become a square in $K_v^\times$ for any $v \in T$, then there exists an $a \in K^\times$ such that*

$$(a,b)_v = \begin{cases} -1 & \text{for } v \in T \\ 1 & \text{otherwise.} \end{cases}$$

Original proof

Source `CFT.pdf`*, physical pages 265–266.*

PROOF. (Following Tate, 1976, 5.2). Let L be the composite of all abelian extensions of K of exponent 2, and let $G = \mathrm{Gal}(L/K)$. By class field theory,

$$G \simeq \boldsymbol{C}_K/2\boldsymbol{C}_K \simeq \mathbb{I}/K^\times \cdot \mathbb{I}^2.$$

The cohomology sequence of

$$0 \longrightarrow \mu_2 \longrightarrow L^\times \xrightarrow{x \mapsto x^2} L^{\times 2} \longrightarrow 0$$

is an exact sequence

$$K^\times \xrightarrow{x \mapsto x^2} K^\times \cap L^{\times 2} \longrightarrow \mathrm{Hom}_{\mathrm{conts}}(G, \mu_2) \longrightarrow 0.$$

Every element of $K^\times$ becomes a square in L, and so we have an isomorphism

$$K^\times/K^{\times 2} \longrightarrow \mathrm{Hom}_{\mathrm{conts}}(\mathbb{I}/K^\times \cdot \mathbb{I}^2, \mu_2).$$

This map sends $a \in K^\times$ to the continuous homomorphism

$$(c_v) \longmapsto \prod (a, c_v)_v$$

(because of the relation between the Hilbert symbol and the local Artin map). Thus, finding a is equivalent to finding a homomorphism $f : \mathbb{I}/\mathbb{I}^2 \longrightarrow \mu_2$ such that

(a) $f = 1$ on principal idèles;

(b)
$$f(1, \dots, 1, i_v(b), 1, \dots, 1) = \begin{cases} -1 & \text{for } v \in T \\ 1 & \text{otherwise.} \end{cases}$$

where i_v is the inclusion $K \hookrightarrow K_v$. For each v, let B_v be the $\mathbb{F}_2$-subspace of $K_v^\times/K_v^{\times 2}$ generated by $i_v(b)$, and let $B = \prod B_v \subset \mathbb{I}/\mathbb{I}^2$. Because $i_v(b)$ is not a square for $v \in T$, there exists a linear form (automatically continuous) $f_1 : B \longrightarrow \mu_2$ satisfying condition (b), and f_1 will extend to a continuous linear form on $\mathbb{I}/\mathbb{I}^2$ satisfying (a) if and only if f_1 takes the value 1 on every principal idéle in B. The value of f_1 on the principal idéle of b is $\prod_{v\in T} -1$, which is 1 because of our assumption that T contains an even number of elements. Let $c \in K^\times$ be such that its idéle lies in B. Then, for every v, $i_v(c) = 1$ or $i_v(b)$ in $K_v^\times/K_v^{\times 2}$. Therefore, $i_v(c)$ becomes a square in $K_v[\sqrt{b}]$ for all v, which (by 1.1) implies that c is a square in $K[\sqrt{b}]$. Hence $c = 1$ or b in $K^\times/K^{\times 2}$, and so f_1 takes the value 1 on its idèle. □

Original supporting excerpts

Source `CFT.pdf`, physical page 238.

THEOREM 1.1 *Let K be a number field containing a primitive nth root of 1. A nonzero element of K is an nth power in K if it is an nth power in K_v for all but possibly finitely many primes v.* PROOF. Recall that, for a field k containing a primitive nth root of 1, a polynomial $X^n - a$ splits completely in $k[X]$ if it has a root k. Let a be a nonzero element of K, and let β be an nth root of a in some extension field. If a is an nth power in K_v for almost all v, then $X^n - a$ splits completely in $K_v[X]$ for almost all v, and so v splits completely in $K[\beta]$. Since this holds for almost all v, (VII, 4.6) shows that $K[\beta] = K$. □

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The nonarchimedean case. Let K be a local field. Recall that the Hilbert symbol $(\cdot,\cdot)$ can defined for $a, b \in K^\times$ by

$$(a,b) = \begin{cases} 1 & \iff X^2 - aY^2 - bZ^2 \text{ represents } 0 \iff aY^2 + bZ^2 \text{ represents } 1 \\ -1 & \text{otherwise.} \end{cases}$$

LEMMA 6.5 *The Hilbert symbol has the following properties:*

(a) *it is bi-multiplicative and $(ac^2, bd^2) = (a,b)$ for all $a, b, c, d \in K^\times$;*

(b) *for any nonsquare $a \in K^\times$, there exists a $b \in K^\times$ such that $(a,b) = -1$;*

(c) $(b,a) = (a,b)^{-1} = (a,b)$;

(d) $(a,-a) = (1,a) = 1$.

PROOF. Obviously, (a,b) does not change when a or b is multiplied by a square. Also, (c) is obvious.

Note that $(a,b) = 1$ if and only if b is a norm from $K[\sqrt{a}]$. From local class field theory, we know that if a is not a square in K, then $\mathrm{Nm}(K[\sqrt{a}]^\times)$ is a subgroup of index 2 in $K^\times$, and therefore $b \longmapsto (a,b)$ is an isomorphism $K^\times/\mathrm{Nm}(K[\sqrt{a}]^\times) \longrightarrow \{\pm 1\}$. This completes the proof of (a) and (b). Finally, $aX^2 - aY^2 = a(X^2 - Y^2)$, and $X^2 - Y^2$ represents a^{-1} because it represents 0. □

The fixed reading for the global reciprocity input, Chapter VII Theorem 8.1, is retained separately as `900008207/r001`, from physical page 230.